

**CrimRxiv**

# **Prediction markets for crime rates: A comprehensive review (as an experiment in collaborating with genAI)**

**genAI Scott Jacques**

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## Foreword, by Scott Jacques

If prediction markets can predict election winners before anyone else, I figure they can be a useful tool for criminologists. But crime rates and prediction markets aren't my expertise. I don't have the time to write a thoughtful paper about their intersection. Instead of writing nothing, I did the next best thing: prompt generative artificial intelligence (genAI) to do so.

This page shares its article, which you should read with great skepticism. I haven't verified anything. I haven't even edited it, except for formatting. The unformatted version is embedded below as a PDF. It also has the prompt I gave to the genAI platform, Perplexity, to create the article.



[Prediction Markets for Crime Rates.pdf](#)

684 KB

As preparation for the final product, I had four conversations with the platform. One was about how to write an article suitable to *Annual Reviews*. The other threads were reviews of predicting crime rates, prediction markets, and prediction markets for crime rates. These conversations are below too. It took me less than an hour to generate them and the article. Was it worth it? [Let me know what you think](#), especially if you have corrections or consider yourself an expert in crime rates, prediction markets, or genAI.



[How to write for Annual Reviews.pdf](#)

793 KB



[Review of predicting crime rates.pdf](#)

653 KB



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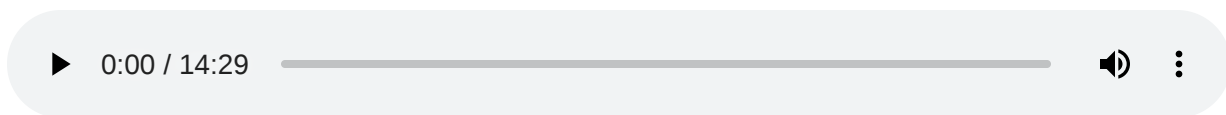
[Review of prediction markets for crime rates.pdf](#)

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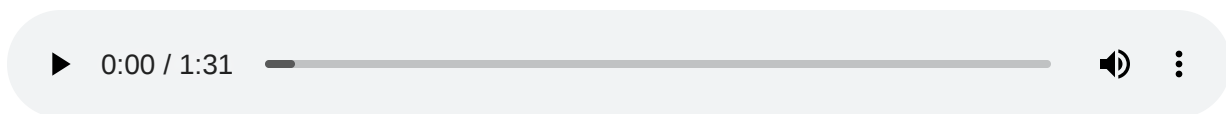
For fun and comprehension, I used another genAI platform, Google NotebookLM, to create supplementary resources. There is a video overview, four audio files (a podcast-style “Deep Dive,” plus a “Brief,” “Critique,” and “Debate”), an infographic, and slide deck. I spent about 30 minutes on them.



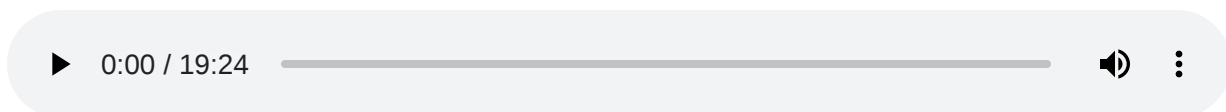
Overview video



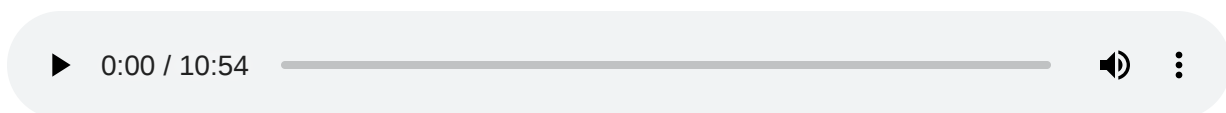
Deep Dive



Brief



Debate



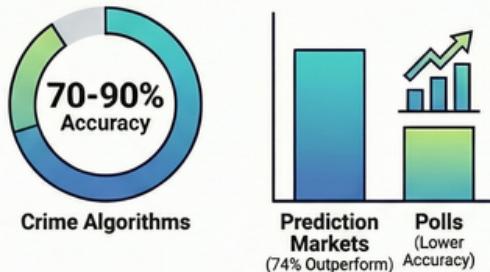
Critique

# Crime Prediction Markets: 15 Years of Unrealized Potential

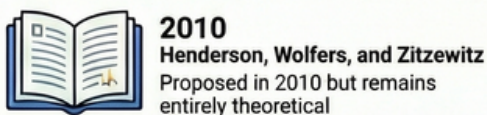
## The Powerful Idea: A 15-Year-Old Proposal



## Two Proven Technologies



## A Foundational Framework Exists



## The Formidable Barriers: Why It Hasn't Happened



### Political & Ethical Backlash

Concept faces criticism for appearing to be "betting on crime" or profiting from tragedy

### Regulatory Hurdles



### Regulatory Hurdles

Legal status is unclear, caught between financial regulation and anti-gambling laws

### Institutional Fragmentation



U.S. has over 17,000 law enforcement agencies, making viable local markets difficult to create

© NotebookLM

Infographic



[Slide deck.pdf](#)

13 MB

In case you're wondering, I wrote this introduction without genAI. It took me longer than the rest combined. My prediction is in the future, criminologists won't have time for traditional authorship. Instead they'll be expected to make non-rhetorical contributions that directly and concretely improve the world. One way is to create, implement, and improve technology—such as prediction-market platforms.

I don't think we should lament a future without authorship as we know it today. The more productive and optimistic approach is to start experimenting with how best to work with genAI. To what end? To increase the quantity, quality, and usage of utilitarian criminology. It'll be interesting to see how long it takes the field to embrace this new approach: at least, to be ok with its occurrence; better, to openly participate in the movement.

There are multiple ways to partake in genAI-authored criminology. For this article, I basically serve as Oz—I told it what to write about. Other roles are coauthor, reader, peer-reviewer, moderator and editor.

I fully expect for genAI articles to be published in peer-review criminology journals. Not for a while, however. In the meantime, I expect to see a lot of anthropocentrism. Good and important research will be rejected simply because it wasn't created by a human.<sup>1</sup> This is discrimination with no place in science. A work should be evaluated on its merit, not based on who or what produced it. If a work advances our knowledge and understanding of empirical phenomena, it should be published so it can be used.

To speed acceptance of genAI-authored articles, it would be good to have a "traditional journal" dedicated to them. My guess is this will be less threatening to human criminologists who want to protect their turf. This would give genAI, and people who use it, space to experiment and push the field forward—instead of being blocked-out by fearful gatekeepers.

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## The Article, by genAI

### Abstract

This review examines the intersection of prediction markets and crime rate forecasting, synthesizing theoretical foundations, methodological approaches, and empirical evidence across three domains: crime prediction, prediction markets generally, and their specific application to criminology. Crime forecasting has evolved from traditional statistical models to sophisticated machine learning approaches, achieving accuracies of 70-90% for hotspot detection. Prediction markets, grounded in Hayekian information aggregation theory and validated across electoral, corporate, and scientific domains, demonstrate superior forecasting performance in diverse contexts. However, the application of prediction markets specifically to crime rates remains almost entirely theoretical, consisting primarily of a 2010 symposium centered on Henderson, Wolfers, and Zitzewitz's foundational proposal. Despite compelling theoretical arguments and demonstrated success in adjacent domains, no crime prediction markets have been implemented in the fifteen years since, due to political toxicity, regulatory uncertainty, institutional fragmentation, and ethical concerns. This review synthesizes current knowledge, identifies critical gaps, and proposes future directions for both academic research and commercial implementation, including specific design considerations for platforms like Polymarket and Kalshi to offer crime rate prediction contracts.

## Introduction

The forecasting of crime rates represents one of the most consequential challenges facing modern criminal justice systems. Accurate predictions enable optimal resource allocation, inform policy decisions, and potentially prevent criminal victimization through proactive interventions. Yet crime forecasting requires aggregating voluminous, dispersed information scattered across incompatible databases, agency jurisdictions, and institutional boundaries—precisely the informational challenge that prediction markets excel at solving. This review examines the unrealized potential of applying market-based forecasting mechanisms to criminology, synthesizing evidence across three interconnected literatures: crime prediction methodologies, prediction market theory and applications, and the nascent scholarship on prediction markets specifically for crime rates.<sup>[1][2][3]</sup>

Prediction markets—market-based mechanisms designed to aggregate dispersed information for forecasting future events—have demonstrated remarkable accuracy across diverse domains. The Iowa Electronic Markets outperformed traditional polls 74% of the time across five presidential elections. Corporate prediction markets at Hewlett-Packard, Google, and Ford achieved forecast error reductions of 25% compared to expert panels. Scientific reproducibility prediction markets correctly predicted 71% of replication outcomes, outperforming individual expert surveys. These successes suggest that prediction markets could enhance crime forecasting by harnessing collective intelligence from police officers, criminologists, community members, and other stakeholders who possess fragments of relevant knowledge.<sup>[2][3]</sup>

Crime prediction research has progressed dramatically over the past two decades, evolving from simple time series models to sophisticated machine learning and deep learning approaches. Extreme Gradient Boosting (XGBoost) and Random Forest models consistently demonstrate accuracies exceeding 90% for certain prediction tasks. Risk Terrain Modeling captures approximately 50% of future crime cases in the top 10% of high-risk cells. Self-exciting point processes outperform traditional hotspot methods for crimes exhibiting contagion effects. Field implementations report crime reductions of 7-41% in targeted areas. Yet these algorithmic approaches face profound challenges around bias, fairness, and accountability, with substantial evidence that historical data reflecting discriminatory policing practices can perpetuate racial inequities through algorithmic feedback loops.<sup>[4][5][6][1]</sup>

The application of prediction markets specifically to crime rates remains almost entirely theoretical. M. Todd Henderson, Justin Wolfers, and Eric Zitzewitz's 2010 article "Predicting Crime" represents the foundational and essentially only comprehensive academic treatment of this intersection. Published in the *Arizona Law Review* alongside four critical commentaries, this symposium generated sophisticated theoretical frameworks, identified formidable implementation barriers, and sparked vigorous debate about ethical implications. Yet fifteen years later, no crime prediction markets have been implemented or empirically tested, leaving the proposal in a state of perpetual potentiality.<sup>[3]</sup>

This review proceeds in four sections. First, we examine the state of crime rate prediction, synthesizing theoretical foundations, methodological approaches, empirical findings, and practical implications. Second, we review prediction market theory, historical development, applications, and comparative performance versus traditional forecasting methods. Third, we analyze the specific literature on prediction markets for crime rates, focusing on the Henderson-Wolfers-Zitzewitz framework and subsequent commentaries. Finally, we synthesize these literatures to propose future directions for academic research and commercial implementation, including specific design considerations for contemporary prediction market platforms.

## Review of Predicting Crime Rates

### Theoretical Foundations

Crime prediction research rests on robust theoretical foundations drawn from environmental criminology and opportunity theories. **Routine Activity Theory**, developed by Lawrence Cohen and Marcus Felson in 1979, posits that crime occurs when three elements converge in space and time: a motivated offender, a suitable target, and the absence of capable guardians. This theory explains spatial-temporal clustering of crime events and informs hotspot prediction by identifying when and where these three elements are most likely to intersect. The theory shifted criminological focus from offender motivation to situational factors, suggesting that crime rates can change without changes in the number of motivated offenders—simply by altering target suitability or guardian presence.<sup>[1]</sup>

**Crime Pattern Theory**, formulated by Paul and Patricia Brantingham in 1993, explains that crime is not randomly distributed but follows patterns shaped by offenders' routine activities and the environmental backcloth. Offenders develop cognitive templates—mental maps of their awareness spaces—based on their daily routines, and crimes tend to cluster along the edges between different land uses and near activity nodes. This theory provides the conceptual foundation for spatial crime forecasting, explaining why certain locations consistently experience elevated crime levels regardless of demographic turnover.<sup>[1]</sup>

**Social Disorganization Theory**, originating with Clifford Shaw and Henry McKay's work in the 1940s, links neighborhood structural characteristics—particularly poverty, residential instability, and racial heterogeneity—to crime rates through weakened informal social control. This theory has been refined through the concept of **Collective Efficacy**, introduced by Robert Sampson, Stephen Raudenbush, and Felton Earls in 1997, which defines communities' capacity to activate social ties to achieve collective goals like crime control. Research demonstrates that collective efficacy mediates much of the relationship between concentrated disadvantage and violence, with high collective efficacy neighborhoods experiencing substantially lower crime rates even when controlling for structural factors.<sup>[1]</sup>

These theoretical frameworks collectively explain why crime is predictable: it follows systematic patterns shaped by opportunity structures, routine activities, and environmental characteristics. This theoretical

coherence provides the foundation for applying statistical and machine learning methods to crime forecasting.  
[1]

## Methodological Approaches

The methodological landscape of crime prediction has evolved dramatically, progressing from traditional statistical models to sophisticated machine learning and deep learning approaches. A systematic review of spatial crime forecasting identified 66 distinct algorithms used across 32 high-quality studies, demonstrating remarkable methodological diversity.<sup>[7][1]</sup>

**Traditional statistical methods** include time series models, particularly Autoregressive Integrated Moving Average (ARIMA) models, which have long been employed for crime forecasting. Studies using ARIMA for crime prediction report accuracy rates of 80-84% for one- to two-year-ahead forecasts. For example, an ARIMA model forecasting London crime achieved 80% accuracy in predicting monthly crime counts two years ahead. These models are particularly effective for long-term strategic planning and budget forecasting but struggle with spatial heterogeneity and abrupt changes in crime patterns.<sup>[1]</sup>

**Machine learning methods** have become dominant in crime prediction due to their superior performance in capturing complex patterns. Decision tree-based algorithms, including individual decision trees and ensemble methods like Random Forest and Gradient Boosting, consistently outperform traditional statistical models. Random Forest demonstrates accuracy rates of 70-90% across various crime prediction tasks. XGBoost (Extreme Gradient Boosting) frequently emerges as the best-performing algorithm, achieving accuracies above 90% in some studies through its sequential tree-building process that corrects errors from previous trees. For instance, XGBoost achieved an R-squared of 0.995 with a mean absolute error of 16.318 in predicting crime in the Netherlands. The algorithm's feature importance outputs, often enhanced with Shapley Additive Explanations (SHAP), provide interpretability—a critical advantage for operational deployment.<sup>[4][1]</sup>

**Deep learning approaches** have emerged to address spatiotemporal crime prediction. Long Short-Term Memory (LSTM) networks excel at capturing long-term temporal dependencies in sequential data, outperforming ARIMA in some comparative studies. Diffusion Convolutional Gated Recurrent Units (DCGRUs), used in models like CrimeForecaster and HAGEN, combine graph neural networks with recurrent architectures to model spatial dependencies between regions and temporal evolution simultaneously. These models represent crime data as graphs where nodes are geographic regions and edges represent spatial relationships, allowing the model to learn how crime spreads across connected areas. ST-SHN (Spatial-Temporal Hypergraph Network) achieved over 27% accuracy in crime occurrence prediction—a 65% improvement over earlier models.<sup>[8][1]</sup>

**Spatial analysis methods** remain fundamental to crime forecasting. Kernel Density Estimation (KDE) remains one of the most popular techniques for hotspot mapping, achieving the highest Prediction Accuracy Index

(PAI) scores among traditional spatial methods. **Risk Terrain Modeling (RTM)** represents an evolution beyond traditional hotspot approaches by identifying environmental risk factors that create opportunities for crime rather than simply mapping where crime has occurred historically. A systematic review and meta-analysis found that RTM captures approximately 50% of future crime cases in the top 10% of high-risk cells—performance substantially better than random allocation. RTM has demonstrated effectiveness across diverse crime types including acquisitive crimes, violent crimes, child maltreatment, terrorism, drug-related crimes, and driving while intoxicated. Field implementations show RTM is roughly 70% better than event-dependent predictions alone, with composite vulnerability-exposure models performing more than 100% better than event-dependent predictions.<sup>[9][7][1]</sup>

**Self-Exciting Point Processes (SEPPs)**, particularly Hawkes processes, model crime as a stochastic process where each event increases the probability of subsequent events in nearby locations and times. Originally developed for earthquake aftershock modeling, SEPPs have been adapted to crime by recognizing that crimes often trigger additional crimes. Studies show that SEPPs outperform traditional hotspot methods for crimes exhibiting strong near-repeat patterns.<sup>[1]</sup>

## Empirical Findings

Empirical research demonstrates that crime prediction models achieve moderate to high accuracy depending on the task, data quality, and methods employed. For **hotspot detection**—the most common prediction task—accuracies range from 70% to 90%. Tree-based ensemble methods (Random Forest, XGBoost) consistently achieve the upper end of this range, while simpler methods like logistic regression and decision trees fall toward the lower end. A 2025 systematic review following PRISMA methodology analyzed 50 articles and confirmed that XGBoost, Random Forest, Gradient Boosting Decision Trees (GBDT), and ARIMA are the most prominent models, with deep learning models like LSTM showing high performance in dynamic urban environments.<sup>[4][1]</sup>

**Crime type prediction** proves more challenging. Neural networks achieved only 16.4% accuracy in predicting the specific type among 27 crime categories, improving to 27.1% when categories were grouped into seven broader classes. This reflects the inherent difficulty of distinguishing among crime types based solely on spatiotemporal and demographic features.<sup>[1]</sup>

**Feature importance** analysis consistently identifies lagged crime rates as the strongest predictors across studies. Demographic variables including population density, age structure, and racial composition show moderate predictive power, though their use raises ethical concerns about potential bias. Socioeconomic factors such as unemployment rates, poverty levels, and education attainment predict property crime more strongly than violent crime. Points of interest (POIs) and urban infrastructure features demonstrate substantial predictive value—the presence of bars, convenience stores, transit stations, and vacant properties correlates with elevated crime risk across multiple studies.<sup>[1]</sup>

**Implementation impacts** from field evaluations yield mixed but generally positive results. The Los Angeles Police Department's randomized controlled trial of PredPol found that the algorithm correctly predicted crime locations in 4.7% of forecasts compared to 2.1% for human crime analysts—more than twice the accuracy. Deploying patrols to algorithmically identified hotspots reduced crime by 7.4% per week, translating to an estimated \$9 million annual savings when accounting for costs to victims, courts, and society.<sup>[1]</sup>

Risk Terrain Modeling implementations show substantial crime reductions: Newark experienced a 35% reduction in gun crimes, Colorado Springs achieved a 33% reduction in vehicle theft, and Glendale reduced robberies by more than 40%. The UK's Grip Programme evaluation found that hot spot policing led to a 41% reduction in hospital admissions for violence with injury in targeted areas. However, not all implementations succeed. A predictive policing intervention in Shreveport, Louisiana found no significant crime reduction differences between treatment and control areas, with evaluators noting that implementation fidelity issues and skepticism among district commanders may have undermined the intervention.<sup>[10][1]</sup>

Importantly, **displacement effects**—the concern that crime prevention in one area simply pushes crime to adjacent areas—appear rare. A comprehensive evaluation of the UK's New Deal for Communities programme found evidence of crime displacement in only 2% of cases where it might occur, while 23% of cases showed diffusion of benefits, where crime reductions spread to surrounding areas beyond the intervention zone. This finding, consistent with previous meta-evaluations, suggests that fears of displacement often exceed its reality.<sup>[1]</sup>

## Bias and Fairness Concerns

A critical empirical finding is the racial bias inherent in many predictive policing systems. ProPublica's investigation of the COMPAS risk assessment algorithm revealed that Black defendants were falsely classified as high-risk at almost twice the rate of White defendants, while White defendants were more frequently mislabeled as low-risk. This bias stems from historical data reflecting disproportionate policing in minority communities, creating feedback loops where predictive models direct more police resources to already over-policed areas, generating more arrests, which further reinforce the predictions.<sup>[5][6][11][1]</sup>

Research demonstrates that algorithmic tools often inherit and amplify systemic inequities. Even when algorithms do not explicitly use race as a feature, proxies such as zip code, prior arrests, and social network data correlate with race, perpetuating discrimination. Legal scholars argue these practices may violate the Equal Protection Clause of the Fourteenth Amendment, as they result in disparate impacts on protected groups even without discriminatory intent. Efforts to mitigate bias include cleansing historical data, incorporating diverse data sources beyond crime reports, and designing algorithms with fairness constraints. However, fundamental tensions exist between predictive accuracy and fairness—optimizing for one often compromises the other.<sup>[11][5][1]</sup>

## Practical Implications

Predictive analytics enable more strategic and efficient allocation of limited law enforcement resources. By identifying high-risk areas and times, agencies can optimize officer scheduling, reduce response times by 15-35%, and deploy personnel where they are most needed. The LAPD's experience demonstrates the fiscal benefits—an estimated \$9 million in annual savings from reduced crime and more efficient operations. Hot spot policing, informed by crime predictions, consistently shows effectiveness in reducing crime by 7-41% in targeted areas without substantial displacement.<sup>[12][1]</sup>

Beyond place-based prediction, risk assessment tools inform decisions about individuals at sentencing, parole, and probation. Machine learning models for predicting criminal recidivism achieve moderate accuracy (AUC around 0.816) when combining clinical risk factors with demographic and behavioral data. However, the predictive performance of risk assessment tools ranges from poor to moderate, with smaller studies systematically overestimating performance. Critical concerns surround person-based risk assessment, particularly feedback loops that reinforce bias and lack of transparency in proprietary algorithms.<sup>[1]</sup>

## Review of Prediction Markets

### Theoretical Foundations and Conceptual Origins

The intellectual foundations of prediction markets rest on several key economic theories that emerged throughout the twentieth century. The concept that markets can efficiently aggregate dispersed information traces back to **Friedrich Hayek's** seminal 1945 article "The Use of Knowledge in Society," in which he argued that prices in competitive markets serve as signals that communicate information about relative scarcity and coordinate economic activity. Hayek observed that no single planner could possess all the knowledge held by dispersed individuals in society, and that the price system functions as a "mechanism for communicating information" that allows for the efficient use of decentralized knowledge. This insight—that prices can encapsulate and transmit information more efficiently than centralized planning—provided the conceptual foundation for using market mechanisms as forecasting tools.<sup>[2]</sup>

The **efficient markets hypothesis (EMH)**, developed and popularized by Eugene Fama in the 1960s and 1970s, provided another crucial theoretical pillar. Fama's influential 1970 paper "Efficient Capital Markets: A Review of Theory and Empirical Work" formalized the hypothesis that in efficient markets, prices "fully reflect" all available information. He distinguished three forms of market efficiency: weak form (past prices cannot predict future performance), semi-strong form (publicly available information is rapidly incorporated), and strong form (even insider information is reflected in prices). The EMH suggested that if markets efficiently aggregate information, then market prices should provide the best available forecasts of future events.<sup>[2]</sup>

Paul Samuelson contributed the mathematical formalization showing that in efficient markets, properly anticipated prices follow a martingale process—meaning future price changes are unpredictable based on

current information. His 1965 proof that "properly anticipated prices fluctuate randomly" demonstrated that if markets efficiently incorporate information, price movements should appear as a random walk relative to what is already known. This theoretical result implied that market prices represent optimal forecasts given available information, providing a mathematical foundation for interpreting prediction market prices as probability estimates.<sup>[2]</sup>

The concept of the "**wisdom of crowds**" provided an additional theoretical rationale for prediction markets. Francis Galton's famous 1906 experiment at a country fair, where 800 people estimated the weight of an ox, demonstrated that the median of crowd estimates (1,207 pounds) came remarkably close to the true weight (1,198 pounds)—within one percent accuracy. This finding, published in 1907, showed that aggregated judgments from diverse groups can be more accurate than individual expert estimates. James Surowiecki popularized this concept in his 2004 book *The Wisdom of Crowds*, arguing that under the right conditions—diversity, independence, decentralization, and effective aggregation mechanisms—groups often make better decisions than any single member.<sup>[2]</sup>

**Experimental economics** provided empirical validation of information aggregation through markets. Charles Plott and Shyam Sunder conducted pioneering laboratory experiments in the early 1980s demonstrating that properly designed markets populated by humans could aggregate dispersed information and converge to rational expectations equilibrium. Their experimental work showed that even when individual traders possess only partial information, market prices can efficiently incorporate the totality of dispersed knowledge. These experiments established the proof-of-concept that markets could serve as information aggregation mechanisms, setting the stage for practical prediction market applications.<sup>[2]</sup>

## Historical Development

The first modern electronic prediction market, the **Iowa Electronic Markets (IEM)**, was launched in 1988 by the Henry B. Tippie College of Business at the University of Iowa. Originally designed as a small-scale, real-money futures market to forecast the 1988 U.S. presidential election, the IEM allowed participants to trade contracts whose payoffs depended on election outcomes. Over more than a dozen years of operation, the IEM consistently demonstrated superior forecasting accuracy. Berg and colleagues found that IEM predictions were closer to actual election results than polls 74 percent of the time across five presidential elections from 1988 to 2004. The market proved particularly effective at long time horizons, significantly outperforming polls when predictions were made more than 100 days before elections.<sup>[2]</sup>

The IEM demonstrated several important properties of prediction markets. First, prices adjusted rapidly to new information, often within minutes of significant news events. Second, the markets exhibited relative efficiency even with small numbers of traders and limited liquidity. Third, despite evidence that many individual traders exhibited behavioral biases—including overconfidence and irrational trading patterns—marginal price-setting traders drove prices toward efficient levels. However, IEM research also revealed systematic biases, including

a transitory overconfidence bias where traders initially overweighted their private information, though this effect diminished as markets matured.<sup>[2]</sup>

Following the IEM's success, political prediction markets proliferated internationally throughout the 1990s and 2000s. **TradeSports and Intrade**, founded in Ireland, became the most prominent commercial platforms. In the 2012 U.S. presidential election, Intrade correctly predicted 49 of 51 jurisdictions (states plus Washington D.C.), missing the final Electoral College tally by just one vote. However, Intrade's success proved short-lived. In 2012, the U.S. Commodity Futures Trading Commission (CFTC) filed a complaint against Intrade for offering commodity options without proper registration. Following an FBI raid and mounting legal pressure, Intrade ceased operations in 2013.<sup>[2]</sup>

**Corporate applications** emerged parallel to public political markets. Kay-Yut Chen and Charles Plott conducted influential research on corporate prediction markets at Hewlett-Packard in the early 2000s. Their results showed that prediction market forecasts outperformed HP's official expert forecasts 75 percent of the time and achieved a 25 percent reduction in forecast error. Google launched extensive internal prediction markets in 2005, allowing all employees to trade on whether quarterly Objectives and Key Results (OKRs) would be achieved. Analysis by Bo Cowgill and Eric Zitzewitz found that Google's markets were well-calibrated and outperformed alternative forecasting methods, though they exhibited systematic biases including optimism about internal projects and overreaction to new information. Ford Motor Company employed prediction markets to forecast weekly vehicle sales, achieving more accurate predictions than expert forecasts with a 25 percent improvement in forecast accuracy.<sup>[2]</sup>

Despite these successes, corporate adoption remained limited. Research identified organizational barriers including hierarchical resistance, power dynamics, and executives' reluctance to democratize decision-making inputs from lower organizational levels. The tension between the decentralized nature of prediction markets and centralized corporate authority structures limited broader adoption.<sup>[2]</sup>

## Methodological Innovations

Significant theoretical and computational work addressed the design of prediction market mechanisms. **Robin Hanson** made foundational contributions in multiple areas. His work on **market scoring rules**—particularly the logarithmic market scoring rule (LMSR)—provided an automated market maker mechanism that ensured liquidity and bounded the market maker's maximum loss. LMSR became the de facto standard for many prediction market implementations because it maintains consistent prices, offers infinite liquidity, and guarantees the market maker's worst-case loss grows only logarithmically with the number of possible outcomes.<sup>[2]</sup>

Hanson also pioneered research on **combinatorial prediction markets**, which allow trading on logical combinations of events. His 2003 work on combinatorial market design enabled markets to reveal not just

marginal probabilities but also conditional probabilities and correlations between events. This innovation dramatically expanded the range of questions prediction markets could address, allowing users to trade on statements like "If candidate A wins, then policy B will be implemented."<sup>[2]</sup>

David Pennock developed **dynamic pari-mutuel market mechanisms** that combined features of traditional parimutuel betting with continuous market trading. His 2004 framework addressed how to design markets that dynamically adjust prices while maintaining coherent probability distributions across multiple outcomes. Research on **Arrow-Debreu securities** provided the standard contract structure for prediction markets—state-contingent contracts that pay a fixed amount if a specific outcome occurs and nothing otherwise, allowing market prices to be directly interpreted as implied probabilities.<sup>[2]</sup>

## Contemporary Developments

**Blockchain-based prediction markets** emerged in the late 2010s. Augur, launched on the Ethereum blockchain in July 2018, became the first major decentralized prediction market platform. Unlike centralized platforms, Augur operates without a single controlling entity, using smart contracts to execute trades and resolve outcomes. Other decentralized platforms including Gnosis and Polymarket followed, leveraging blockchain technology to offer global access, reduced costs, and censorship resistance.<sup>[2]</sup>

As of 2025, prediction markets are experiencing renewed attention following their performance in electoral forecasting and the expansion of legal prediction market platforms. **Polymarket** has grown to become the largest prediction market platform, recording over \$18 billion in trading volume throughout 2024 and 2025. The platform operates on the Polygon blockchain, offering low transaction fees and fast settlement times, with users trading exclusively with USDC. Polymarket is reportedly in preliminary discussions to secure new funding, aiming for a valuation between \$12 billion and \$15 billion—a remarkable tenfold increase from its valuation just four months prior.<sup>[13][14][15][16]</sup>

**Kalshi**, the regulated competitor, recently completed a \$1 billion funding round, bringing its valuation to \$11 billion. Operating as a federally regulated exchange under CFTC oversight, Kalshi has experienced rapid growth over the past year, reportedly reaching an annualized trading volume of \$50 billion in October 2025, a significant increase from around \$300 million the previous year. Both Kalshi and Polymarket have successfully predicted outcomes ranging from political elections to economic indicators.<sup>[14][15][17]</sup>

## Comparative Accuracy

Empirical research demonstrates that prediction markets generally outperform alternative forecasting methods. A 2020 systematic review and meta-analysis found that prediction markets were on average 79 percent more accurate than alternative forecasting methods. The IEM outperformed traditional polls 74% of the time across five presidential elections. Corporate prediction markets at HP, Google, and Ford achieved forecast error reductions of 25-75% compared to expert forecasts.<sup>[2]</sup>

Scientific reproducibility prediction markets conducted by Anna Dreber, Colin Camerer, and colleagues correctly predicted 71 percent of replication outcomes, with market prices highly correlated with actual replication success. The prediction markets outperformed surveys of individual expert forecasts, demonstrating their value for aggregating peer beliefs about research credibility. Economic indicator prediction markets launched by Goldman Sachs and Deutsche Bank performed comparably to consensus forecasts from professional economists, while also revealing information about forecast uncertainty through the distribution of contract prices.<sup>[2]</sup>

However, research also documented **systematic biases and market failures**. The favorite-longshot bias—where longshots are overbet and favorites underbet—appeared in some prediction markets, though evidence was mixed. Overconfidence bias led traders to overweight private information and underestimate uncertainty, particularly early in market operation. Optimism bias appeared particularly pronounced in corporate prediction markets, where employees systematically overestimated the probability of favorable internal outcomes. Experimental studies found that markets could fall into "information traps" where prices failed to aggregate information, instead exhibiting cascades where traders imitated others rather than trading on private information.<sup>[2]</sup>

## Regulatory Challenges

The legal and regulatory status of prediction markets generated substantial controversy. In the United States, the CFTC's authority over commodity futures raised questions about whether prediction markets constituted commodity options, securities, or gambling under various federal and state laws. The 2003 cancellation of DARPA's Policy Analysis Market (PAM)—proposed by Robin Hanson to use prediction markets for forecasting geopolitical events including terrorism—sparked intense debate. Critics characterized it as a "terrorism futures market" and congressional opposition led to immediate termination despite support from researchers who argued markets could provide valuable intelligence insights.<sup>[2]</sup>

The CFTC historically prohibited prediction markets on political elections and similar events, arguing they lacked "economic purpose" and were "contrary to the public interest". However, the commission granted limited exemptions for academic research. In 2020, **Kalshi** received CFTC approval as a designated contract market, allowing it to offer event contracts on outcomes with "economic consequences". Subsequent litigation between Kalshi and the CFTC over the permissibility of political election contracts resulted in court rulings favoring Kalshi, opening the door for regulated prediction markets on previously prohibited topics.<sup>[18][2]</sup>

State regulators raised concerns about prediction markets potentially operating as unlicensed gambling or sports betting operations. The 2024-2025 period saw increased scrutiny as prediction markets experienced explosive growth around U.S. elections, with gaming commissions in New York and other states issuing cease-and-desist orders and threatening to revoke licenses of operators participating in prediction markets. Arizona's gaming director formally urged the CFTC to take action against sports prediction markets, arguing that these

platforms effectively operate as sportsbooks without appropriate oversight and sidestep state gambling laws.<sup>[18]</sup>  
[2]

## Review of Prediction Markets for Crime Rates

### The Henderson-Wolfers-Zitzewitz Framework

The academic literature on prediction markets applied to criminology and crime forecasting is remarkably sparse, consisting primarily of a single seminal article and a cluster of critical commentaries published in a 2010 symposium. The intellectual genesis of prediction markets for crime rates can be traced to a 2008 working paper by M. Todd Henderson (University of Chicago Law School), Justin Wolfers (University of Pennsylvania Wharton School), and Eric Zitzewitz (Dartmouth College), which was subsequently published in refined form in the *Arizona Law Review* in 2010.<sup>[3]</sup>

Henderson, Wolfers, and Zitzewitz grounded their proposal in both theoretical foundations and empirical evidence from adjacent domains. Drawing on Friedrich Hayek's insights about price mechanisms as information aggregators and the emerging literature on prediction markets in elections, business forecasting, and public health, the authors identified crime as "perhaps the most obvious policy area" for prediction market application, yet one that had been entirely overlooked. The theoretical framework rests on several key premises. First, information relevant to crime forecasting is voluminous, dispersed, and difficult to process efficiently. Police officers on patrol, criminologists with statistical models, neighborhood residents with local knowledge, and potentially even offenders themselves all possess fragments of relevant information that existing institutional structures fail to aggregate systematically. Second, current crime forecasting methods are "extremely crude"—relying on gut instinct, backward-looking mapping tools, or models that are "unproven, difficult to interpret, and expensive." Third, prediction markets have demonstrated superior performance compared to expert panels, polls, and econometric models across multiple domains.<sup>[3]</sup>

The Henderson-Wolfers-Zitzewitz framework makes two primary **theoretical contributions** beyond simply proposing crime prediction markets. First, they developed **prediction market event studies** designed to estimate the causal impact of real or hypothetical policy changes on crime rates. Given the enormous costs and risks of implementing major criminal justice reforms—such as capital punishment moratoriums or changes to mandatory minimum sentences—without knowledge of their effects, these event studies would allow policymakers to gauge market participants' collective wisdom about policy impacts before implementation. The market would react to policy announcements or simulated scenarios, revealing embedded expectations about crime effects.<sup>[3]</sup>

Second, and more technically sophisticated, the authors addressed the **policy feedback loop problem**. Predictions about future crime rates inherently embed assumptions about policy responses to those predictions, creating a circular dependency. If a market predicts rising burglary rates, policymakers may respond by

increasing police patrols, which would then affect actual burglary rates, invalidating the original prediction. Henderson, Wolfers, and Zitzewitz developed a formal method for "unpacking" market forecasts to isolate underlying crime factors from embedded policy response expectations, allowing policymakers to interpret market prices even when they reflect endogenous policy adjustments.<sup>[3]</sup>

The authors also addressed **practical design considerations** including contract structure, market liquidity requirements, participant incentives (financial and reputational), and the optimal geographic and temporal scope of markets. They concluded with a "modest proposal" for federal experimentation: a market predicting Uniform Crime Reports (UCR) statistics over three years, involving approximately 200 expert participants (criminologists and police chiefs) endowed with \$200 each, with modest cash-out thresholds and non-financial rewards for top performers.<sup>[3]</sup>

## The 2010 Arizona Law Review Symposium

The publication of "Predicting Crime" in the Arizona Law Review (Volume 52, Issue 1, 2010) was accompanied by four critical commentaries that examined practical, ethical, institutional, and political dimensions of the proposal. This symposium represents the most concentrated scholarly engagement with prediction markets for criminology to date.<sup>[3]</sup>

**Miriam A. Cherry** (University of the Pacific, McGeorge School of Law) authored "Morality and Markets: A Comment on Predicting Crime," which examined the normative and political obstacles to implementing crime prediction markets. Cherry's analysis was informed by the cautionary tale of the DARPA Policy Analysis Market (PAM), a terrorism prediction market proposed after September 11, 2001, that was canceled before launch following fierce political criticism. Senators condemned PAM on moral grounds, arguing that "betting" on terrorist attacks was problematic and that such markets could enable terrorists to profit from their own attacks.<sup>[3]</sup>

Cherry argued that similar ethical concerns would plague crime prediction markets: the perception of profiting from crime predictions, especially if criminals themselves participated, raised fundamental questions about market morality. She situated these concerns within the broader "commodification" and "taboo trades" literature, noting parallels to controversial markets for organ donation and child adoption. To address these concerns, Cherry proposed several design modifications. First, limit participation to experts or government employees rather than opening markets to the general public, which would eliminate concerns about criminals profiting from insider knowledge. Second, remove financial incentives entirely, replacing them with reputational rewards and renaming the mechanism "prediction sharing technology" rather than a "market" to reduce the appearance of gambling on crime. While these changes might reduce market efficiency by eliminating the "local perspective" of everyday citizens and potentially reducing expert participation, Cherry argued they would enhance political feasibility.<sup>[3]</sup>

**Thomas E. Feucht** (National Institute of Justice) and **William J. Sabol** (Bureau of Justice Statistics) provided the most skeptical commentary in "Comment on a 'Modest Proposal' for a Crime Prediction Market". As federal officials who would potentially be responsible for implementing the proposed experiment, they raised fundamental questions about scale, participation, incentives, and policy utility.<sup>[3]</sup>

On **scale and scope**, Feucht and Sabol challenged the authors' focus on national UCR crime statistics, arguing that "crime is local". State and local jurisdictions dominate criminal justice spending and decision-making—in 2005, local jurisdictions spent \$103.77 billion on criminal justice compared to only \$35.41 billion by the federal government. A national UCR prediction market would produce information with limited policy relevance for local decisionmakers who control resource allocation, policing strategies, and prosecutorial priorities. Conversely, local prediction markets face severe challenges: small participant pools, easily-manipulated local crime data, insufficient expert traders in any single jurisdiction, and vulnerability to affiliation bias when one group dominates the market.<sup>[3]</sup>

Most fundamentally, Feucht and Sabol questioned the **policy utility** of crime prediction markets. They observed that across many prediction market applications, the mechanism serves primarily as a game (election betting) or winner-identification device (flu forecasting) rather than directly informing decisions. Accuracy is insufficient; information must translate into action. Local decisionmakers face political risks in relying on forecasts from anonymous market participants rather than named experts—if predictions prove wrong, accountability becomes diffuse and justification becomes difficult. The authors concluded that "the basic question of the relevance of market predictions for crime policy and decision-making may seem to be a concern that is secondary to finding out simply whether or not a market can predict crime in the first place. We think that because crime prediction markets would be new to the field, it would be important to confront the 'so what' question".<sup>[3]</sup>

**Ronald F. Wright** (Wake Forest University School of Law) contributed "Fragmented Users of Crime Predictions," which explored how the extreme decentralization of American criminal justice institutions would affect prediction market utility. Wright identified 2,344 separate state prosecutors' offices and 17,876 state and local law enforcement agencies in the United States, with 74% employing fewer than 24 officers.<sup>[3]</sup>

Wright argued that prediction markets would prove most valuable for **state-level centralized users** with dispersed information needs, such as sentencing commissions forecasting prison populations, state funding authorities allocating resources among jurisdictions, or corrections departments evaluating recidivism programs. These actors face the coordination challenge that prediction markets excel at solving: synthesizing information from "many diverse sources of information" scattered across incompatible databases, jurisdictions, and time periods. Markets could "leapfrog an entire generation of data compatibility problems and speed up the centralization of criminal justice policy."<sup>[3]</sup>

However, **local users** in fragmented jurisdictions face "thin markets" unlikely to generate reliable predictions. Small police departments and prosecutors' offices would depend on bids from "only a handful of officers or observers," making forecasts vulnerable to idiosyncratic biases and manipulation. Yet Wright expressed moral reservations about case-level use of market predictions for sentencing, noting that "we expect someone—generally the judge—to take individual responsibility for sentencing choices". Criminal justice decisions involve not merely factual prediction but also "accountable moral evaluation." Wright concluded that "users must not confuse a component from a holistic judgment. The unaccountable market prediction of the future must remain distinct from the accountable moral evaluation of the present".<sup>[3]</sup>

## Subsequent Developments and the Absence of Implementation

Following the 2010 symposium, academic engagement with prediction markets specifically for crime forecasting has been virtually nonexistent. Comprehensive searches identified no subsequent peer-reviewed articles, working papers, or books dedicated to developing, refining, or empirically testing the concept between 2010 and 2025. This absence is particularly striking given several contextual factors. First, prediction markets generally have proliferated during this period, with platforms like Kalshi, Polymarket, and PredictIt expanding into new domains and gaining mainstream attention. Second, criminal justice has experienced a data revolution, with predictive policing algorithms, risk assessment tools, and machine learning models becoming widespread. Third, forecasting generally has become more prominent in criminal justice policy discussions. Yet these parallel developments have remained distinct from prediction markets.<sup>[3]</sup>

Several factors explain why prediction markets for crime have failed to progress beyond the 2010 symposium.

**Political and ethical toxicity:** The PAM precedent demonstrated that government-sponsored prediction markets on sensitive topics face severe political headwinds. The association with "betting on crime" creates public relations vulnerabilities that risk-averse government agencies would naturally avoid. **Legal constraints:** The Unlawful Internet Gambling Enforcement Act of 2006 created a "chilly" legal environment for real-money prediction markets in the United States. **Institutional fragmentation:** Wright's analysis proved prescient—the extreme decentralization of American criminal justice means the most natural users (local police and prosecutors) operate in markets too thin to generate reliable predictions. **Implementation complexity:** Feucht and Sabol's concerns about design parameters remain unresolved without empirical trials. **Alternative technological paths:** The rise of big data analytics and machine learning may have offered a less politically controversial path to improved crime forecasting. **Absence of empirical validation:** With no implemented crime prediction markets, the theoretical claims about superior forecasting accuracy remain untested in the criminological context.<sup>[3]</sup>

## Summary Synthesis and Future Directions

## Synthesis of Key Insights

This review has examined three interconnected literatures—crime prediction methodologies, prediction market theory and applications, and prediction markets specifically for crime rates—revealing both remarkable progress and profound unrealized potential. Crime forecasting has evolved from crude statistical models to sophisticated machine learning approaches achieving 70-90% accuracy for hotspot detection and enabling crime reductions of 7-41% through targeted interventions. Yet these algorithmic systems face fundamental challenges around bias, fairness, and accountability, with substantial evidence that historical data reflecting discriminatory policing can perpetuate racial inequities through feedback loops.<sup>[6][5][4][1]</sup>

Prediction markets, grounded in Hayekian information aggregation theory and validated across electoral, corporate, and scientific domains, demonstrate superior forecasting performance—outperforming traditional methods by 25-79% on average. Markets efficiently aggregate dispersed information from diverse participants, adjust rapidly to new information, and provide probabilistic forecasts with quantified uncertainty. Contemporary platforms like Polymarket and Kalshi have achieved multibillion-dollar valuations and trading volumes, demonstrating commercial viability and mainstream acceptance.<sup>[15][13][14][2]</sup>

Yet the application of prediction markets specifically to crime rates remains almost entirely theoretical, consisting of Henderson, Wolfers, and Zitzewitz's 2010 proposal and accompanying commentaries. Despite compelling theoretical arguments, sophisticated methodological frameworks for event studies and policy feedback loop decomposition, and demonstrated success in adjacent domains, no crime prediction markets have been implemented in fifteen years. The literature identifies formidable barriers: political toxicity from associations with "betting on crime," legal uncertainty around gambling versus derivatives classification, institutional fragmentation creating thin markets for local jurisdictions, and fundamental questions about policy utility when accountability requires named experts rather than anonymous market participants.<sup>[3]</sup>

Several **key tensions** emerge from synthesizing these literatures. First, while algorithmic crime prediction faces criticism for perpetuating bias through historical data, prediction markets aggregate human judgments that may equally reflect or even amplify societal prejudices. If market participants hold biased beliefs about which neighborhoods will experience crime, markets will efficiently aggregate those biases into prices. Second, while transparency is frequently cited as essential for algorithmic accountability, many advanced prediction market mechanisms (particularly automated market makers and combinatorial markets) operate as complex "black boxes" potentially less interpretable than simpler statistical models. Third, while decentralization is celebrated as prediction markets' strength—aggregating dispersed local knowledge—the extreme decentralization of American criminal justice may paradoxically undermine market functionality by fragmenting potential participants across too many jurisdictions.

## Future Directions: Academia and Criminology

Academic research on prediction markets for crime rates faces fundamental limitations: without empirical implementation, theoretical debates cannot advance beyond speculation. The field requires a **proof-of-concept demonstration** to establish whether markets can forecast crime with acceptable accuracy in real-world settings. Several research priorities emerge:

**Experimental trials:** The most urgent need is implementing Henderson, Wolfers, and Zitzewitz's "modest proposal" or variations thereof. A federally-sponsored prediction market forecasting national UCR statistics would provide essential baseline evidence about market performance, participant composition, trading dynamics, and forecast accuracy. While Feucht and Sabol correctly note that national statistics have limited local policy relevance, establishing that markets can forecast crime at any level would justify further experimentation. Alternative experimental designs could target state-level outcomes (prison populations, statewide crime rates) where centralized users exist and jurisdictional coverage is manageable.<sup>[3]</sup>

**Methodological refinement:** Research should develop and test contract structures optimized for crime forecasting. Standard binary contracts ("Will homicides exceed X?") could be compared with continuous contracts (estimating the exact homicide count), combinatorial contracts (conditional probabilities like "If unemployment rises, will property crime increase?"), and composite indices aggregating multiple crime types. Market scoring rule parameters should be optimized to balance liquidity provision with bounded loss. Temporal horizons require investigation—are markets more accurate forecasting one month, one quarter, or one year ahead? Geographic granularity must be calibrated to participant knowledge and data availability.<sup>[2][3]</sup>

**Comparative evaluation:** Rigorous comparisons between prediction market forecasts and alternative methods are essential. Markets should be benchmarked against statistical models (ARIMA), machine learning algorithms (XGBoost, Random Forest), deep learning approaches (LSTM, graph neural networks), expert panels, and institutional forecasts from police departments and criminologists. Evaluation metrics should include not just point accuracy but calibration, resolution, Brier scores, and measures of forecast uncertainty. The relative performance across different crime types (violent versus property, common versus rare) would reveal markets' comparative advantages.<sup>[4][1][2]</sup>

**Bias and fairness assessment:** Given the profound ethical concerns around algorithmic bias in predictive policing, research must examine whether prediction markets exhibit systematic racial, geographic, or socioeconomic biases. Do markets disproportionately predict crime in minority communities regardless of actual incidence? Can bias mitigation techniques developed for algorithms (fairness constraints, bias-aware training) be adapted to market designs (participant diversity requirements, weighted voting schemes)? How do markets' biases compare to those of algorithmic systems and human experts?<sup>[5][6][1]</sup>

**Theoretical development:** The policy feedback loop problem identified by Henderson, Wolfers, and Zitzewitz requires deeper theoretical and empirical investigation. Formal models should characterize equilibria when market prices influence policy responses that affect outcomes being predicted. Experimental economics studies could test whether participants can effectively "unpack" policy-contingent forecasts. Game-theoretic analysis should examine strategic behavior when participants have interests in outcomes (e.g., prosecutors who benefit politically from appearing tough on crime).<sup>[3]</sup>

**Participant composition and incentives:** Research should identify optimal participant pools. Should markets be restricted to experts (criminologists, police chiefs) or opened to general publics? How does participant diversity affect accuracy? What incentive structures—financial rewards, reputational scoring, gamification—most effectively motivate participation and honest revelation of beliefs? Cherry's proposal to eliminate financial incentives and rebrand as "prediction sharing technology" merits empirical testing against traditional market designs.<sup>[3]</sup>

**Ethical governance frameworks:** Interdisciplinary research combining criminology, law, ethics, and computer science should develop governance frameworks for crime prediction markets. What disclosure requirements should apply? How can insider trading concerns be addressed when police officers possess non-public operational information? What community oversight mechanisms would be appropriate? How should markets handle morally sensitive outcomes like mass shootings or terrorism?

## Future Directions: Commercial Prediction Markets

Contemporary prediction market platforms—particularly Polymarket and Kalshi—have achieved remarkable scale and sophistication, offering contracts on diverse domains from elections and economics to sports and entertainment. The 2024-2025 period witnessed unprecedented growth, with these platforms processing billions of dollars in trading volume and attracting mainstream media attention. This commercial success creates opportunities to implement crime rate prediction markets that academic institutions and government agencies have been unable or unwilling to pursue.<sup>[17][16][13][14][15]</sup>

## Implementation Example: US Monthly Homicide Rates

A concrete implementation pathway would involve Kalshi or Polymarket offering event contracts on **US monthly homicide rates**, building on existing FBI crime data infrastructure. The FBI's Crime Data Explorer provides monthly homicide data through the National Incident-Based Reporting System (NIBRS), though participation and reporting completeness present challenges. Despite data gaps—with approximately one-third of law enforcement agencies missing from 2022 federal data—improved participation (6,000 more agencies reporting in 2022 versus 2021) suggests feasibility.<sup>[19][20]</sup>

**Contract structure** could follow several models. **Binary contracts** would offer simple yes/no propositions: "Will US homicides in January 2026 exceed 2,000?" with contracts resolving to \$1 if true, \$0 if false. Market

prices directly represent implied probabilities (a 65-cent price implies 65% probability). **Range contracts** would segment outcomes into bins: "Homicides in January 2026: <1,800 (A), 1,800-2,000 (B), 2,000-2,200 (C), >2,200 (D)" with one contract paying \$1 and others \$0 based on actual outcome. This structure provides more granular probability distributions. **Over/under contracts** would establish a threshold with continuous payoffs proportional to distance from threshold, similar to sports betting spreads.

**Data sources** would primarily utilize FBI Uniform Crime Reports (UCR) monthly data and Supplementary Homicide Reports (SHR), supplemented by Real-Time Crime Index data from AH Datalytics which audits and corrects for underreporting. Contract resolution would occur 60-90 days after month-end to allow reporting stabilization—the FBI's monthly data exhibits substantial undercounting immediately after month-end that corrects over subsequent weeks. Resolution criteria must specify how to handle revised data; the most conservative approach would use FBI official figures released at a predetermined date (e.g., 90 days post-month) regardless of subsequent revisions.<sup>[19]</sup>

**Technical implementation** for **Kalshi** would leverage its existing CFTC-regulated infrastructure as a designated contract market. Kalshi operates with full KYC/AML compliance, accepts USD deposits through traditional banking rails, and provides institutional-grade settlement and clearing. Homicide contracts would be categorized under "Economics" or a new "Public Safety" category. The platform's existing risk management systems—position limits, circuit breakers, manipulation monitoring—would apply. Kalshi's regulatory compliance framework addresses state gambling concerns, though some state regulators have challenged event contracts as unauthorized gambling.<sup>[21][14][15][18]</sup>

For **Polymarket**, implementation would utilize its Polygon-based blockchain infrastructure with USDC settlement. The decentralized architecture offers global access (though US participation remains restricted pending regulatory clarity) and lower transaction costs. Polymarket's outcome resolution relies on UMA's decentralized oracle system where token holders vote on outcomes, introducing additional complexity for crime data that may be subject to revision. The platform would need to specify exact data sources (FBI CDE API endpoints) and resolution timing to minimize disputes.<sup>[16][22]</sup>

**Participant composition** would likely include criminologists and researchers studying homicide trends, law enforcement professionals with operational insights into violence patterns, data journalists tracking crime statistics, risk analysts at insurance companies concerned about violence exposure, social scientists studying structural determinants of homicide, and general public with interests in public safety policy. Unlike local crime markets that might attract insider trading from officers with operational knowledge, national aggregate statistics reduce informational advantages from privileged access. However, researchers with advance access to CDC mortality data or state-level statistics could potentially trade ahead of FBI releases.

**Liquidity provision** represents a critical challenge. Kalshi's automated market maker using logarithmic market scoring rules (LMSR) ensures continuous liquidity with bounded loss. Initial market-making would require

capital allocation proportional to outcome uncertainty—more volatile homicide predictions demand larger liquidity pools. Polymarket similarly employs LMSR variants to maintain continuous trading. Empirically, prediction market liquidity correlates with public interest; homicide contracts might generate modest volume initially, requiring aggressive market-making to attract participants.<sup>[2]</sup>

**Ethical and regulatory considerations** demand careful attention. **Moral concerns** about "profiting from murder" directly parallel the DARPA PAM controversy that doomed terrorism prediction markets. Platforms could address this through:<sup>[3]</sup>

- **Charitable donation mechanisms:** Kalshi or Polymarket could commit portions of trading fees to violence prevention nonprofits, gun safety organizations, or victim support services.
- **Framing and messaging:** Emphasizing the public good of accurate forecasting for resource allocation rather than "betting" or "gambling" language.
- **Participant restrictions:** Limiting markets to institutional and researcher accounts rather than general public might reduce sensationalism, though this would sacrifice diversity and liquidity.

**Regulatory pathways** differ between platforms. Kalshi's CFTC designation provides regulatory clarity but subjects the platform to extensive compliance requirements and potential CFTC objections to specific contracts. The commission could argue that homicide contracts lack "economic purpose" or are "contrary to public interest"—the same reasoning historically used to prohibit political contracts. However, Kalshi's 2024 legal victory over the CFTC regarding political contracts establishes precedent that event contracts with significant public interest serve legitimate purposes.<sup>[23][18]</sup>

State regulators pose perhaps the greater threat. Gaming commissions in Arizona, Pennsylvania, and New York have challenged prediction markets as unlicensed gambling operations. Homicide contracts could face similar scrutiny, particularly if marketed to general consumers rather than sophisticated traders. Platforms must prepare legal defenses grounded in the derivatives/gambling distinction: prediction markets aggregate information for forecasting rather than offering games of chance for entertainment.<sup>[18]</sup>

**Implementation timeline** could proceed as follows:

- **Months 1-3:** Platform develops contract specifications, builds data ingestion APIs from FBI Crime Data Explorer, establishes resolution procedures including dispute protocols for data revisions, and conducts legal review and regulatory consultation.
- **Months 4-6:** Soft launch with limited contracts (perhaps quarterly rather than monthly initially to reduce resolution frequency), targeted outreach to criminology researchers and policy analysts, and evaluation of trading volume, liquidity, and forecast accuracy versus baselines.
- **Months 7-12:** Expansion to monthly contracts if initial performance proves successful, introduction of additional crime types (violent crime index, property crime, specific offenses), geographic segmentation

(state-level contracts for high-population states), and public communication of forecasting performance to build credibility.

**Use cases and beneficiaries** extend beyond intellectual curiosity. **Municipal governments and police departments** could use market forecasts to inform budget planning, resource allocation, and officer deployment scheduling. **Insurance companies** pricing policies with violence exclusions could hedge exposure through crime rate contracts. **Social service organizations** delivering violence prevention programs could calibrate intervention intensity based on forecasted trends. **Researchers** studying crime dynamics could access market-aggregated expectations to benchmark statistical models and identify periods when actual outcomes diverge from collective expectations, signaling structural breaks or policy impacts.

**Accuracy evaluation** must compare market forecasts against alternatives. Baseline comparisons could include:

- **Naive forecasts:** Assuming next month's homicides equal this month's (random walk).
- **Seasonal averages:** Using historical monthly averages accounting for seasonal patterns.
- **ARIMA models:** Time series forecasts from autoregressive models trained on historical data.<sup>[1]</sup>
- **Expert surveys:** Polling criminologists for point estimates and confidence intervals.

Accuracy metrics should include mean absolute error (MAE), root mean squared error (RMSE), and Brier scores for probabilistic calibration. The market would demonstrate value if it achieves forecast errors at least 10-20% lower than naive baselines, though even parity with sophisticated statistical models would justify continued operation given markets' additional benefits (revealing forecast uncertainty, updating continuously with new information, aggregating diverse perspectives).

**Alternative contract designs** could enhance policy relevance. **Conditional contracts** might ask: "If a major gun control law passes, will homicides decrease?" providing event study functionality that Henderson, Wolfers, and Zitzewitz identified as particularly valuable. **Geographic contracts** could offer state-level or city-level homicide forecasts for jurisdictions with reliable reporting. **Demographic contracts** might forecast racial disparities in victimization, though this raises profound ethical concerns about potentially commodifying racial violence. **Cause-specific contracts** could distinguish gun homicides from other methods, providing targeted information for firearm policy debates.<sup>[3]</sup>

**Scalability beyond homicide** would extend to other crimes with reliable national reporting. **Violent crime index** (aggregating murder, rape, robbery, assault) provides a broader indicator. **Property crime** (burglary, larceny, motor vehicle theft) affects larger populations and may generate more trading interest from insurance and retail sectors. **Specific offenses** like carjacking or fentanyl overdoses (technically not crimes but closely related to drug markets) could serve niche forecasting needs. **Recidivism rates** for released prisoners could inform corrections policy and sentencing guidelines.

**Comparison with algorithmic forecasting** reveals complementary strengths. Machine learning models excel at processing vast datasets, identifying complex nonlinear patterns, and producing granular spatial predictions. But algorithms require extensive data infrastructure, suffer from bias embedded in training data, and provide limited interpretability. Prediction markets aggregate human judgment without requiring centralized data collection, potentially incorporate qualitative information (e.g., perceptions of social unrest, political changes) that algorithms miss, and update fluidly as participants revise beliefs. The optimal approach may combine both: algorithmic forecasts could serve as informed priors that market trading refines through incorporation of diverse perspectives and real-time information.<sup>[5][4][1][2]</sup>

## Barriers to Adoption and Mitigation Strategies

- Despite these implementation pathways, substantial barriers remain. **Political sensitivity** surrounding "crime markets" creates reputational risks for platforms. Mitigation strategies include:
- **Academic partnerships:** Collaborating with universities (Iowa Electronic Markets model) to frame markets as research tools rather than commercial gambling.<sup>[2]</sup>
- **Transparency and public education:** Publishing detailed whitepapers explaining information aggregation benefits, forecasting performance, and ethical safeguards.
- **Stakeholder engagement:** Consulting with law enforcement, criminology professional associations, and civil rights organizations before launch to address concerns proactively.
- **Legal uncertainty** particularly at state level requires robust compliance infrastructure. Platforms should:
- **Seek declaratory relief:** Preemptively filing actions in key jurisdictions to establish legal status before state enforcement actions.
- **Maintain regulatory dialogue:** Ongoing communication with CFTC (for Kalshi) and state gaming commissions to address concerns as they arise.
- **Design contracts conservatively:** Avoiding morally controversial outcomes (e.g., individual murder cases, mass shooting events) that could trigger political backlash.
- **Data quality concerns** from incomplete FBI reporting necessitate:
- **Multiple data source triangulation:** Incorporating Real-Time Crime Index, CDC mortality data, and state-level statistics where available.<sup>[19]</sup>
- **Delayed resolution with revision protocols:** Specifying that contracts resolve 90 days post-period using final revised FBI figures.
- **Transparent methodology:** Publishing exact data sources, API endpoints, and calculation methods to enable independent verification.
- **Thin markets and low liquidity** may undermine forecast accuracy. Strategies to enhance participation include:
- **Aggressive market-making:** Committing substantial platform capital to provide continuous liquidity and narrow bid-ask spreads.<sup>[2]</sup>
- **Reduced fees:** Waiving or minimizing transaction fees for crime contracts to encourage trading.

- **Educational content:** Publishing analysis, historical trends, and expert commentary to attract informed participants.
- **Integration with related markets:** Cross-listing crime contracts alongside correlated markets (unemployment, gun sales, political polarization) to attract traders with diverse information.

## Conclusion

This review has synthesized three interconnected literatures to assess the unrealized potential of prediction markets for crime rate forecasting. Crime prediction has evolved into a sophisticated domain combining theoretical foundations from environmental criminology with advanced methodological approaches achieving remarkable forecasting accuracy. Prediction markets have demonstrated superior performance across diverse domains, from elections and corporate forecasting to scientific reproducibility, grounded in robust theoretical frameworks and empirical validation. Yet the application of prediction markets to crime rates remains almost entirely theoretical, consisting of Henderson, Wolfers, and Zitzewitz's 2010 proposal and accompanying commentaries, with no implementations in fifteen years despite compelling theoretical arguments.

The literature reveals both profound promise and formidable barriers. Promise stems from prediction markets' demonstrated ability to aggregate dispersed information more efficiently than centralized expert panels, their continuous updating as new information arrives, and their potential to provide policy-relevant forecasts through event study designs. Barriers include political toxicity from associations with "betting on crime," regulatory uncertainty around gambling versus derivatives classification, institutional fragmentation creating thin markets for local jurisdictions, ethical concerns about commodifying criminal victimization, and fundamental questions about whether anonymous market forecasts can substitute for accountable expert judgment in policy decisions.

Contemporary developments create new opportunities. Commercial prediction market platforms like Polymarket and Kalshi have achieved multibillion-dollar valuations and mainstream acceptance, demonstrating commercial viability at scale. Improved FBI crime data infrastructure through monthly NIBRS reporting provides more timely and granular information than previously available. The regulatory landscape, while complex, has shifted toward greater acceptance of event contracts following Kalshi's legal victories over the CFTC. Machine learning approaches to crime forecasting have matured sufficiently to provide rigorous accuracy baselines against which markets can be evaluated.

Future progress requires coordinated action across academic, governmental, and commercial sectors. Academic criminology must move beyond theoretical speculation to empirical testing, implementing proof-of-concept trials that establish whether markets can forecast crime with acceptable accuracy. Federal agencies like the National Institute of Justice should fund experimental markets following Henderson, Wolfers, and Zitzewitz's modest proposal, accepting that initial failures would provide valuable learning rather than definitive refutation. Commercial platforms should cautiously explore crime rate contracts, beginning with

national aggregate statistics (homicides, violent crime indices) that minimize moral sensitivities while establishing forecasting capabilities.

The fundamental question remains whether prediction markets can serve as tools for positive social transformation—enabling more efficient resource allocation, better-informed policy decisions, and ultimately safer communities—or whether the political, ethical, and practical barriers will continue to consign the idea to the realm of unrealized theoretical innovations. Fifteen years after Henderson, Wolfers, and Zitzewitz's foundational proposal, the time has come to move from speculation to implementation, from theoretical promise to empirical testing. Only through actual deployment can we determine whether markets can help predict and prevent crime, or whether this represents a case where theoretical elegance fails to translate into practical utility. The infrastructure, platforms, and knowledge exist; what remains is the institutional will to experiment, the ethical wisdom to proceed carefully, and the scientific commitment to evaluate rigorously.

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## Footnotes

1. The human-collaborators will be ignored, criticized, even stigmatized. ↵